

E-COMMERCE CONVERSION RATE ANALYSIS

A Data Science Project Using Python, Machine Learning and Power BI

Submitted by:

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B.Tech – Data Science

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E-Commerce Analytics and Prediction

Tools Used:

Python | Google Colab | Scikit-learn | Power BI

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1. INTRODUCTION

E-commerce has become an important part of modern business. Online shopping websites receive a large number of visitors every day, but not every visitor makes a purchase. The percentage of website sessions that result in a purchase is known as the conversion rate.

Understanding visitor behavior is important for e-commerce businesses because it can help them identify which factors influence purchasing decisions.

This project focuses on analyzing e-commerce website visitor behavior using data analysis, visualization, machine learning and business intelligence techniques.

Python is used for data cleaning, exploratory data analysis and machine learning. Power BI is used to create an interactive dashboard that presents important conversion-related insights in an easy-to-understand format.

2. PROBLEM STATEMENT

E-commerce websites receive visitors with different browsing behaviors, traffic sources, visitor types and levels of interaction with website pages. However, only a small proportion of visitors may complete a purchase.

The objective of this project is to analyze website visitor session data and identify the factors associated with successful conversions. The project also aims to develop

machine learning models that can predict whether a website session is likely to result in a purchase.

3. OBJECTIVES

The main objectives of this project are:

1. To analyze e-commerce website visitor behavior.
2. To clean and prepare the dataset for analysis.
3. To calculate the overall e-commerce conversion rate.
4. To compare converted and non-converted website sessions.
5. To identify important factors influencing conversion.
6. To perform exploratory data analysis and visualization.
7. To build machine learning models for conversion prediction.
8. To evaluate the performance of the prediction models.
9. To create an interactive Power BI dashboard.
10. To present the findings in a clear and understandable manner.

4. DATASET DESCRIPTION

The project uses the Online Shoppers Purchasing Intention Dataset. The dataset contains information about online shopping sessions and visitor behavior.

The dataset contains approximately 12,330 website sessions and 18 attributes.

Important attributes include:

Attribute	Description
Administrative	Number of administrative pages visited
Administrative_Duration	Time spent on administrative pages
Informational	Number of informational pages visited
Informational_Duration	Time spent on informational pages
ProductRelated	Number of product-related pages visited
ProductRelated_Duration	Time spent on product-related pages
BounceRates	Percentage of visitors who leave after viewing a page

Attribute	Description
ExitRates	Percentage of exits from pages
PageValues	Average value of pages visited
Month	Month of the session
OperatingSystems	Operating system used
Browser	Browser used
Region	Visitor region
TrafficType	Source/type of traffic
VisitorType	Type of visitor
Weekend	Whether the session occurred on a weekend
Revenue	Whether the session resulted in a purchase

The Revenue attribute is used as the target variable for conversion analysis and prediction.

5. TECHNOLOGIES USED

Python

Used for data loading, cleaning, analysis and machine learning.

Pandas

Used for data manipulation and analysis.

NumPy

Used for numerical operations.

Matplotlib

Used for creating visualizations.

Seaborn

Used for statistical visualizations and correlation analysis.

Scikit-learn

Used for machine learning, data preprocessing and model evaluation.

Google Colab

Used as the development environment for Python analysis.

Power BI

Used to create the interactive e-commerce conversion rate dashboard.

6. METHODOLOGY

The project was completed through the following stages:

Step 1 – Data Collection

The Online Shoppers Purchasing Intention Dataset was obtained and loaded into Google Colab.

Step 2 – Data Understanding

The dataset structure, columns, data types, number of rows and statistical information were examined.

Step 3 – Data Cleaning

The dataset was checked for missing values, duplicate records and incorrect data types.

Step 4 – Exploratory Data Analysis

Visitor behavior was analyzed using statistical summaries and visualizations.

Step 5 – Conversion Rate Analysis

The percentage of sessions that resulted in a purchase was calculated.

Step 6 – Machine Learning

Machine learning models were trained to predict whether a session would result in a purchase.

Step 7 – Model Evaluation

The models were evaluated using accuracy, precision, recall, F1-score and confusion matrix.

Step 8 – Dashboard Creation

An interactive Power BI dashboard was created to present

the major findings.

7. DATA CLEANING

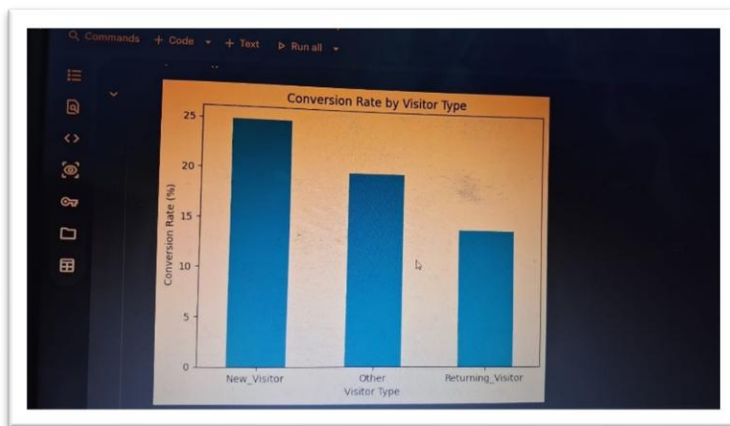
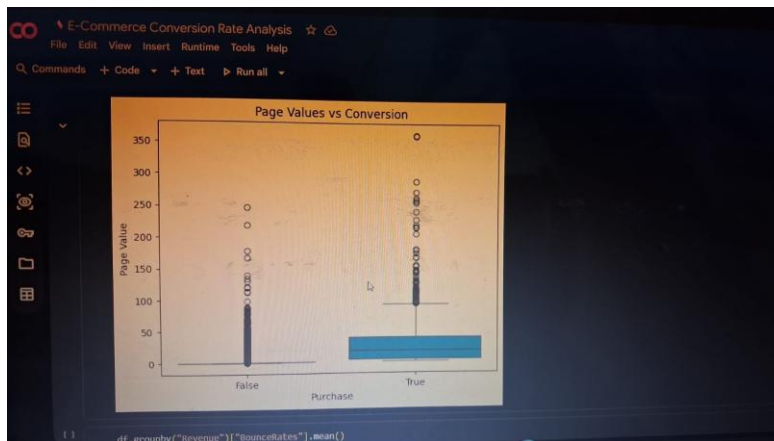
The dataset was loaded into Google Colab using Pandas. The initial dataset was examined using functions such as `head()`, `shape()`, `info()` and `describe()`.

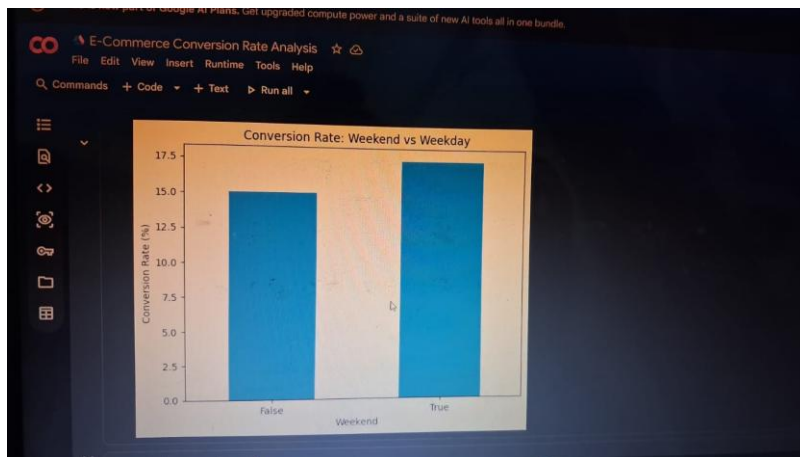
The dataset was checked for:

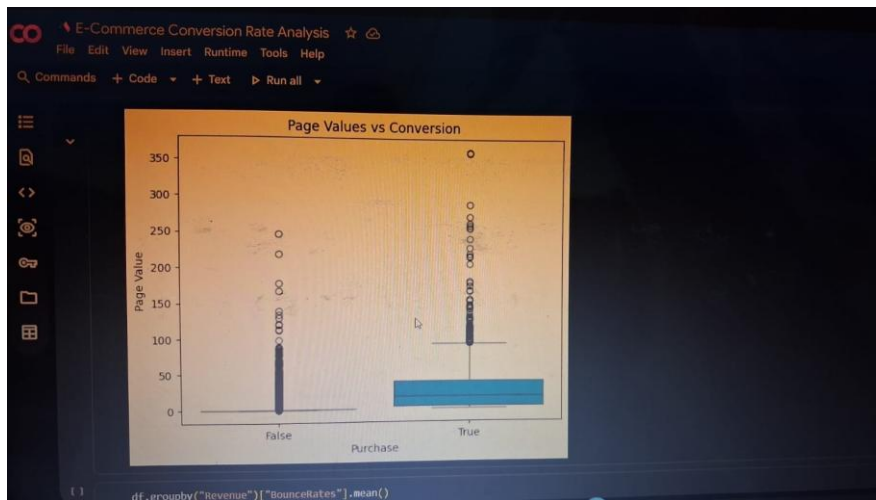
- Missing values
- Duplicate records
- Incorrect data types
- Target variable distribution

Duplicate records were identified and removed where necessary. The categorical variables were converted into an appropriate format for machine learning.

The Revenue column was used as the target variable, where a successful purchase represents a converted session and no purchase represents a non-converted session.







10. MACHINE LEARNING

Machine learning was used to predict whether an e-commerce website session would result in a purchase.

The Revenue column was selected as the target variable.

The dataset was divided into training and testing datasets. Categorical variables were converted into numerical representations using one-hot encoding.

Two machine learning algorithms were used:

10.1 Logistic Regression

Logistic Regression was used as a baseline classification model for predicting whether a session would result in a purchase.

10.2 Random Forest

Random Forest was used as a second classification model. It combines multiple decision trees and can capture complex relationships between the input features and the target variable.

The performance of both models was compared using classification metrics.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression				
Random Forest				

12. POWER BI DASHBOARD

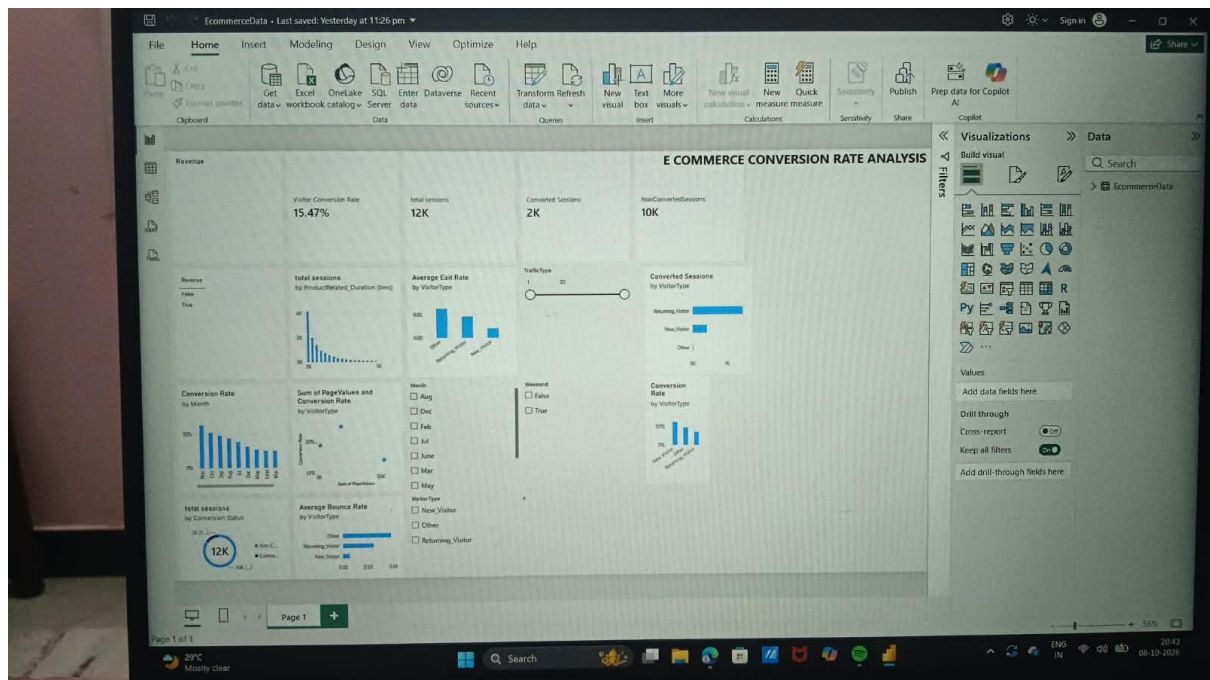
An interactive Power BI dashboard was created to present the e-commerce conversion analysis.

The dashboard contains the following key performance indicators and visualizations:

- Total Sessions
- Converted Sessions
- Conversion Rate
- Non-Converted Sessions
- Conversion Rate by Visitor Type
- Conversion Rate by Month
- Conversion Rate by Traffic Type
- Converted vs Not Converted Sessions
- Conversion Rate by Product Pages Viewed
- Average Page Value by Conversion Status

Interactive slicers were also added for Month, Visitor Type, Traffic Type and Weekend.

The dashboard allows users to filter the data and examine conversion behavior from different perspectives.



13. KEY FINDINGS

The analysis provides several useful insights into e-commerce visitor behavior.

1. The overall conversion rate was calculated using the number of converted sessions compared with total sessions.
2. Different visitor types showed different conversion behaviors.

3. The number of product-related pages visited was analyzed to understand its relationship with conversion.
4. Page Value showed a noticeable relationship with purchasing behavior.
5. Traffic type was compared to identify differences in conversion performance.
6. Monthly conversion patterns were analyzed to identify periods with higher or lower conversion rates.
7. Machine learning models were used to identify whether session-level information could be used to predict conversion.
8. The Power BI dashboard provides an interactive way to explore these patterns.

CONCLUSION

9. The E-Commerce Conversion Rate Analysis project demonstrates how data science and business intelligence techniques can be used to understand online shopping behavior.
10. The project involved data cleaning, exploratory data analysis, visualization, conversion rate analysis and machine learning.
11. Python was used to analyze the dataset and develop prediction models, while Power BI was used to create an interactive dashboard.
12. The analysis helps identify patterns in visitor behavior and factors associated with successful conversions. These

insights can help e-commerce businesses understand their customers and make data-driven decisions to improve website conversion performance.

15. FUTURE SCOPE

The project can be further improved in several ways:

- Use larger and more recent e-commerce datasets.
- Develop more advanced machine learning models.
- Perform hyperparameter tuning.
- Build a real-time conversion monitoring dashboard.
- Develop customer segmentation models.
- Use advanced prediction techniques to estimate purchase probability.
- Integrate the model into an e-commerce website.
- Analyze customer behavior over multiple sessions.

16. REFERENCES

1. Online Shoppers Purchasing Intention Dataset, UCI Machine Learning Repository.
2. Python Documentation.
3. Pandas Documentation.
4. Scikit-learn Documentation.
5. Microsoft Power BI Documentation.

